Exp NO: 4
Date: 4/2/2026

VERIFICATION OF THEVENIN'S THEOREM

AIM:

To verify Thevenin's Theorem practically and and theoretically for the given DC circuit.

APPARATUS REQUIRED:

SL NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated power supply (RPS)	(0-30 V)	1
2	voltmeter	(0-30V) MC	1
3	Ammeter	(0-10 mA) MC	1
4	Resistor	470 Ω 560 Ω 1 K Ω	2 1 1
5.	Bread board	- - -	1
6.	Multimeter	- - -	1

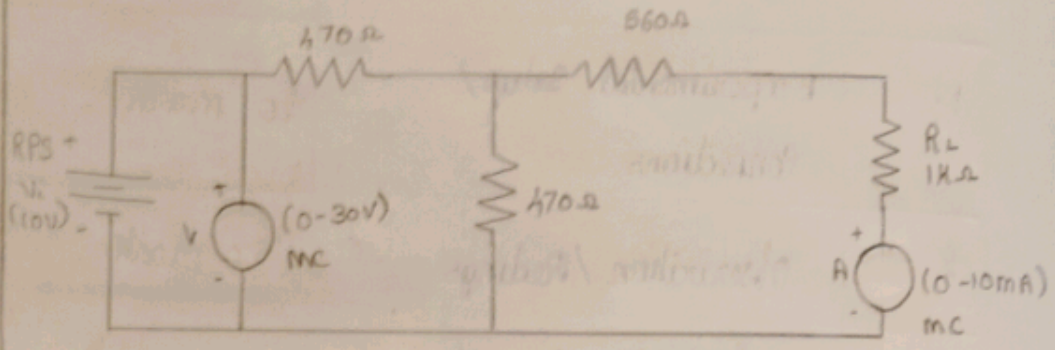
THEORY:

Thevenin's Theorem:

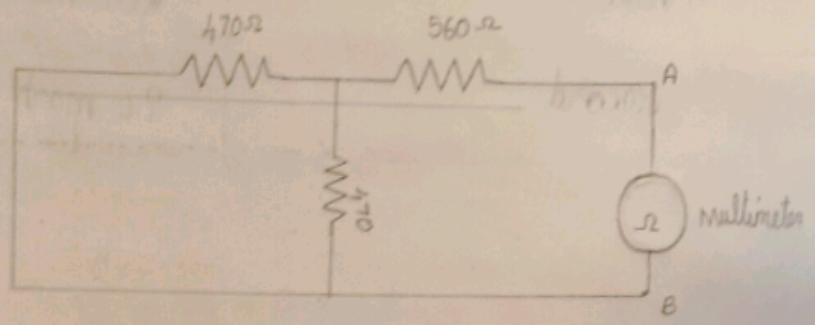
Thevenin's theorem states that 'a linear two terminal circuit can be replaced by an equivalent circuit consisting of a voltage source,

CIRCUIT DIAGRAM: VERIFICATION OF THEVENIN'S THEOREM:

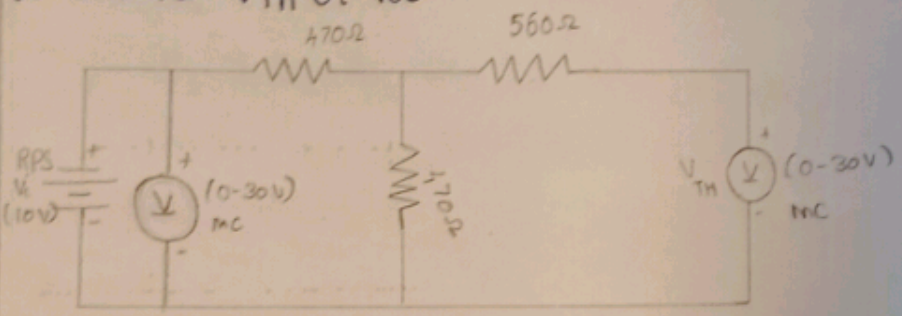
To measure I_L



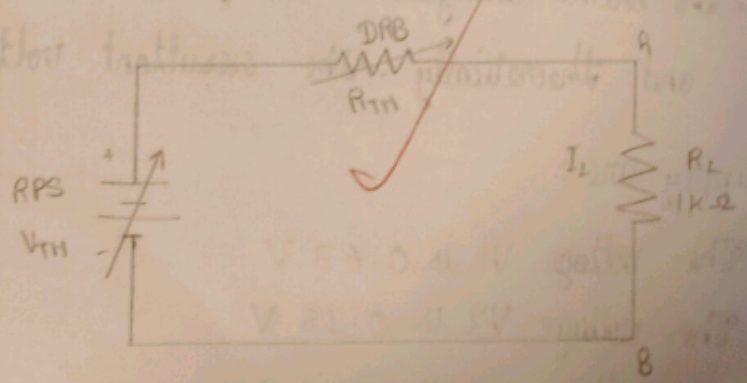
To measure R_{TH} or R_N



To measure V_{TH} or V_{OC}



Thevenin's equivalent circuit



V_{TH} (= V_{OC}) in series with a resistor R_{TH}, where V_{TH} is the open-circuit voltage at the load terminals and R_{TH} is the input or equivalent resistance at the load terminals when the independent sources are turned off.

Thevenins voltage, V_{TH}:

It is the open-circuit voltage across the load terminal (AB) i.e. $V_{TH} = V_{OC}$

Thevenins Resistance, R_{TH}:

It is the look-back resistance across the load terminals when all the sources are replaced by their internal resistances. An ideal voltage source is replaced by short-circuiting as its internal source is replaced by open-circuiting as its internal resistance is infinity.

PROCEDURE:

1. Make the connections as per the circuit diagram 1.
2. Vary the RPS and set an input voltage of 10V.
3. Note down the voltmeter reading (V_i) and ammeter reading (I_L) in tabular column 1.
4. Switch off the supply and make connections for circuit diagram 2.
5. Measure the Thevenins resistance R_{TH} = Norton's resistance R_N.
6. Switch off the supply and make connections for circuit diagram: 3
7. Switch off the supply and make connections for circuit diagram: 4.

TABULAR COLUMN : 1

To measure I_L

V_i (volts)	V_L (amps).
10 V	2.8mA

Tabular column : 2

To measure R_{TH} or R_N

R_{TH} (Ω)	R_N (Ω)
795 Ω	795 Ω

Tabular Column : 3

To measure V_{TH} or V_{OC}

V_i (volts)	V_{TH} (volts).
10V	5 V

MARKS SPLIT-UP :

Sl NO	Description	Mark Alloted	Marks obtained
1.	Experimental setup / Connections	20 Marks	20
2.	Observation / Readings	20 marks	18
3.	Calculations / Results	20 marks	18
4.	Viva	20 marks	18
5.	Record	20 marks	16
		Total	90

RESULT :

Thus Thevenin's theorem is verified practically and theoretically